

Tirth Shah

PYTHON & AI/ML DEVELOPER

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🔗 Portfolio Website

🌐 LinkedIn Profile

🐙 GitHub Profile

PROFESSIONAL SUMMARY

- Python and AI/ML Engineer focused on the design, optimization, and deployment of high-performance intelligent systems.
- Expertise in full-stack web development and scalable AWS serverless architecture, emphasizing automation and end-to-end efficiency.
- Skilled in NLP, Large Language Models (LLMs), and LLM integration, engineering complex RAG-based systems, chatbots, and summarization pipelines.
- Experienced in end-to-end ML system development — from data preprocessing and model training to REST API deployment on AWS Lambda — with hands-on production exposure to LangChain, RAG pipelines, and real-time data processing.

PROFESSIONAL EXPERIENCE

Python & AI/ML Developer

Dec 2024 - Present

NicheTech Computer Solutions | Ahmedabad, Gujarat

- Engineered enterprise-grade RAG and summarization pipelines for PDF document corpora using LangChain, Large Language Models (LLMs), FAISS, and GPT-4, increasing operational user engagement by 40%.
- Developed a Django-based Enterprise Suite for Sandesh, integrating project management (PMS), training (LMS), and resource modules (HRMS) using PostgreSQL to optimize manual administration workflow times by 30% for 200+ internal users.
- Architected Trip Teller (AI travel assistant) and KeywordsClub (SEO content tool) using FastAPI and LangChain with OpenAI/Groq APIs, serving 5,000+ monthly queries with sub-2s response latency.
- Automated critical AWS data pipelines via Lambda and DynamoDB, slashing processing times by 95%.

Product Manager Intern

Jun 2024 - Aug 2024

Tally Prime | Remote

- Automated 500+ invoices via Excel scripting, accelerating billing preparation cycles by 60%.

TECHNICAL PROJECTS

Digi Bank — Django

Django, React, SQLite, REST APIs

Developed a secure Django & React banking application featuring JWT session authentication, database-level locking for concurrency, and validation APIs achieving 100% double-entry bookkeeping audit accuracy.

Fraud Detection Pipeline

XGBoost, Scikit-Learn, SMOTE, Pandas

Built a fraud detection pipeline using XGBoost and scikit-learn on imbalanced transaction data; achieved 97.3% AUC-ROC with SMOTE-based oversampling, processing 10K+ records.

Serverless Face Recognition

AWS Lambda, Rekognition, S3, Python

Implemented an AWS serverless architecture utilizing Lambda and Rekognition for real-time, low-latency facial identification upon S3 object uploads.

Smart Agriculture System

Random Forest, Python, Scikit-Learn, Pandas

Developed a predictive soil analysis classification engine using Random Forest models to recommend crops based on environmental sensor datasets.

Stream-Together watch party platform

Python, WebSockets, Aply API

Synchronized YouTube watch party platform featuring low-latency live chat and stream sync pipelines managed via Aply.

E-Med CLI Pharmacy Management

Python, SQLite

Developed a Python-based database CLI tool optimized for tracking pharmacy inventory transactions, order processing, and customer query storage.

TECHNICAL SKILLS

Languages & Frameworks: Python, Golang, Django, FastAPI, Flask, React, Node.js, JavaScript, HTML, CSS, REST APIs

Databases & Tools: PostgreSQL, MongoDB, DynamoDB, Docker, Git, n8n, Pandas, NumPy, GitHub Actions (CI/CD basics)

AI, ML & LLM: Large Language Models (LLMs), LangChain, OpenAI API, Groq API, RAG, NLP, Prompt Engineering, Vector Databases (FAISS), Hugging Face, AWS (Lambda, S3, DynamoDB, Rekognition), Generative AI, IoT, N8N

EDUCATION

B.Tech in Computer Science

Nov 2022 - April 2026

L.J. Institute Of Information and Technology, Ahmedabad, India

CGPA: 9.0 / 10.0

CERTIFICATIONS

✓ Claude 101 - Anthropic

✓ ML Specialization - Google

✓ GenAI-Powered Apps - IBM

✓ AWS Certified Cloud Practitioner

✓ Python Data Science & AI - IBM